

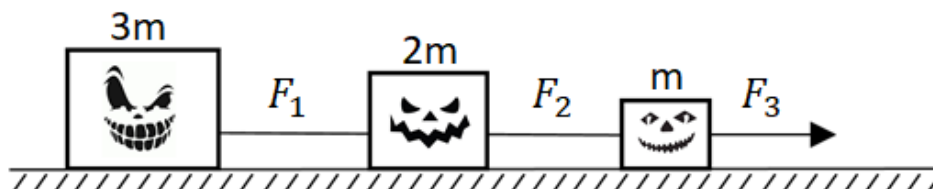
Part 1: Multiple Choice - Choose the answers that best answers the questions below. (4 pts each)

1. Two objects, A and B, move toward one another. Object A has twice the mass and half the speed of object B. Which of the following describes the forces the objects exert on each other when they collide and provides the best explanation?

- A) The force exerted by A on B will be twice as great as the force exerted by B on A, because A has the twice the mass of B.
- B) The force exerted by A on B will be half as great as the force exerted by B on A, because A has half the speed of B.
- C) The forces exerted by each object on the other are the same, because the B will have an equal reaction force to A's force on B.
- D) The force exerted by A on B will be half as great as the force exerted by B on A, because B will have a greater acceleration.

2. For which of the following objects will be the net force acting on it be zero?

- A) A football is thrown straight up into the air.
- B) A mine cart traveling uphill and slowing down until it rolls back down the hill.
- C) A hamburger in free-fall.
- D) A crate being pushed uphill at constant speed.



3. Which of the following expressions **cannot** be used to represent the acceleration of the spooky system above?

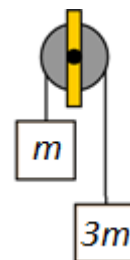
- A) $F_3/6m$
- B) $F_2/2m$
- C) $F_1/3m$
- D) All of the above could be used to represent the acceleration of the system

4. A student with a mass of 50 kg is standing on a bathroom scale while riding an elevator. No one is quite sure what the student is trying to accomplish, but the reading on the scale is 400 N. Which of the following is a correct description of the elevator's motion?

- A) Moving upward with increasing speed
- B) Moving downward with constant speed
- C) Moving downward with increasing speed
- D) Moving downward with decreasing speed

5. Consider the pulley system shown. What is the magnitude of the acceleration of the blocks.

- A) $g/2$
- B) $2g/3$
- C) $g/3$
- D) g



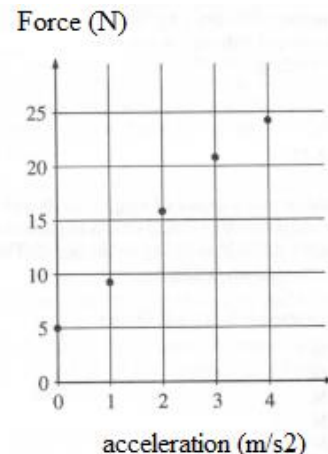
6. A spooky block is placed on an incline as shown. The block travels down the incline. Suppose the angle of the incline is increased from θ to 2θ . Which of the following statements is FALSE:



- A) The component of the block's weight acting downward parallel to the incline increases.
- B) The component of the block's weight acting directly perpendicular into to the incline increases.
- C) The acceleration of the block increases.
- D) The block's total weight stays the same.

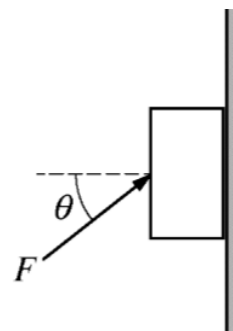
7. The net force on an object vs. its acceleration is shown in the graph to the right. Which of the following quantities is closest to the mass of the object?

- A) 3 kg
- B) 5 kg
- C) 7 kg
- D) 10 kg



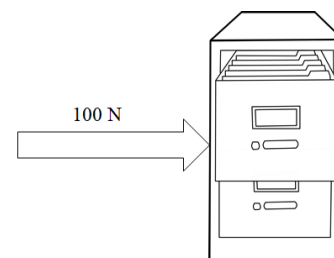
8. A block is held at rest against a wall by a force of magnitude F exerted at an angle θ from the horizontal, as shown. Let W be the weight of the block, F_N be the normal force exerted by the wall on the block, and F_f be the frictional force exerted by the wall on the block. Which of the following statements is true regarding the magnitudes of these forces?

- A) $F = W/\sin\theta$
- B) $F_N = W + F\sin\theta$
- C) $F = W + F_N - F_f$
- D) $F_N = F\cos\theta$



9. A 24 kg file cabinet is initially at rest on a rough surface. The coefficient of static friction between the cabinet and ground is $\mu_s = 0.50$ and the coefficient of kinetic friction is $\mu_k = 0.30$. The cabinet is pushed by a 100 N. Find the magnitude of the force of friction.

- A) 100 N
- B) 120 N
- C) 72 N
- D) 20 N



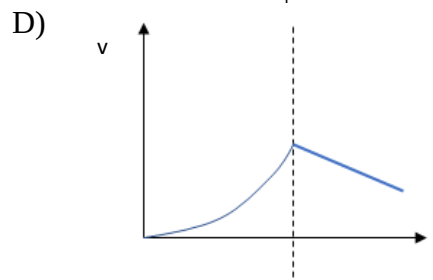
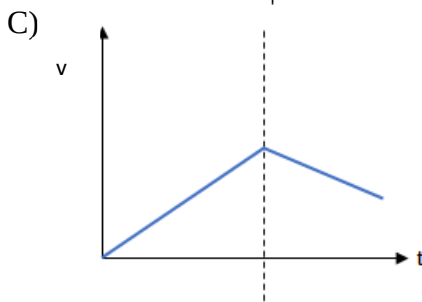
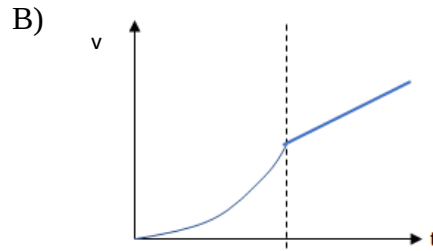
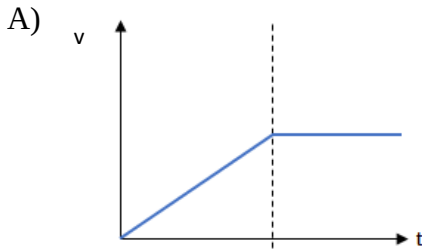
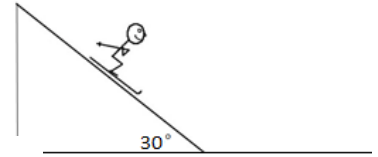
10. Scotty Skeletor the Spooky Skier skies spookily and swiftly down a spooky snowy slope at constant speed. The coefficient of friction between the skier and the incline is .36. Find the angle of inclination.

- A) 16.7°
- B) 21.1°
- C) 19.8°
- D) 68.9°

11. A large suitcase is moved across the dirty floor at Miami International Airport. The frictional force of the floor on a large suitcase is least when the suitcase is:

- A) dragged by a force parallel to the floor.
- B) pulled by a force directed at an angle θ above the floor.
- C) pushed by a force directed at an angle θ into the floor.
- D) turned on its side and pushed by a force parallel to the floor.

12. A skier slides down an incline that makes an angle of 30° with the horizontal. At the bottom of the incline, the skier continues on a horizontal surface. All surfaces are **frictionless**. Which of the following graphs could show the speed skier as a function of time? (dashed line represents leaving the incline).



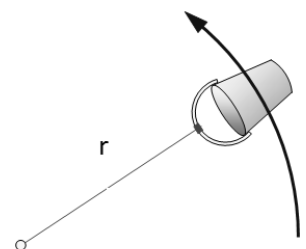
13. A spooky object moves in a spooky horizontal circle at constant speed. In what direction is the object's spooky acceleration?

- A) There is no acceleration of the object since it has constant speed.
- B) Outwards directly away from the center of the circle of motion.
- C) Towards the center of the circle of motion.
- D) Tangent to the circle of motion at the object's location.

14. Some dude is whirling a thing on a string in a horizontal circle at constant speed. The string makes one complete revolution in two seconds; and the magnitude of the tension in the string is F . The dude then speeds up the stone, keeping the radius of the circle unchanged, so that the string makes one complete revolutions every second. What happens to the tension in the string?

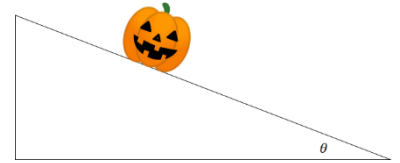
- A) The magnitude of the tension increases to four times its original value, $4F$.
- B) The magnitude of the tension increases to twice its original value, $2F$.
- C) The magnitude of the tension reduces to half of its original value, $F/2$.
- D) The magnitude of the tension reduces to one-fourth of its original value, $F/4$.

15. Max whirls a 0.5 kg bucket of Mountain Dew at the end of a string of length $r = 0.3 \text{ m}$ in a vertical circle. At the top of the path, the bucket's speed is the minimum value such that the Mountain Dew does not fall out of the bucket. What is the tension in the string at this point?

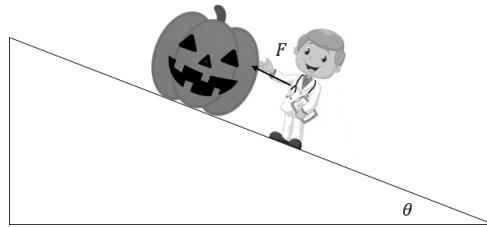


- A) 10 N
- B) 2.2 N
- C) 1.7 N
- D) 0 N

16. (15 points) A Jack-o-Latern of mass M kg is placed on a ramp that is inclined at θ with the horizontal as shown. The coefficient of static friction between the block and incline is μ_s and the coefficient of kinetic friction is μ_k .



- a) The jack-o-latern is placed on the incline at rest. Determine an expression for the force of static friction

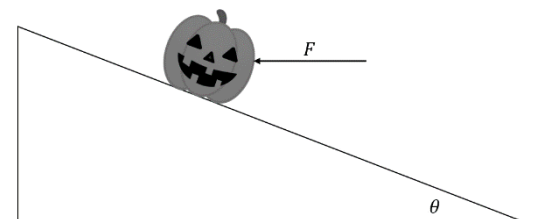


- b) Dr. Acula now pushes the jack-o-latern up the incline as shown. He applies a force F parallel to the incline.
- On the diagram below, draw a free-body diagram of all forces acting on the jack-o-latern as it is pushed uphill.

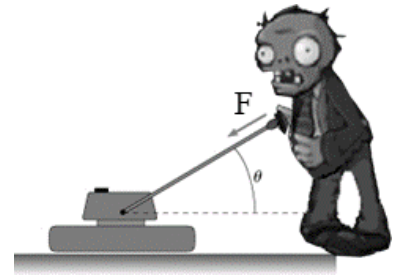


- Derive an expression for the acceleration of the jack-o-latern up the incline. Show your work.

- Suppose the force F is now applied parallel to the ground instead of parallel to the incline as shown. Would the magnitude of the acceleration now be greater than, less than, or equal to the acceleration from ii? Justify your answer.



17. (15 points) A spooky zombie applies a force of magnitude $F = 140 \text{ N}$ at angle of $\theta = 30^\circ$ below the horizontal on a lawn-mower of mass 12.0 across a horizontal surface. Treat the lawn-mower as a sliding object with the coefficient of kinetic friction between the mower and ground being $\mu = 0.20$.



- a) Draw a free body diagram of all forces acting on the mower.



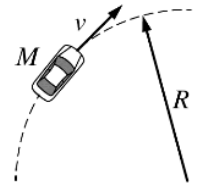
- b) Is the normal force between the lawn mover and the ground greater than, less than, or equal to the weight of the lawn mower?

_____greater than _____less than _____equal to

- c) Calculate the acceleration of the mower. Show your work.

- d) Suppose that the lawn-mower reaches a speed of 15 m/s , and the zombie lets go. Calculate the time required for the mower to stop after being let go.

18. (13 points) a) A car of mass $M=920$ kg travels at constant speed in circular path of radius $R = 25$ m. The car completes 5 complete cycles in 25 seconds.



i. Calculate the minimum coefficient of friction needed to keep the car in a circle.

ii. Suppose the friction between the car's tire and roads suddenly drops to 0. Briefly describe the subsequent motion of the car.

b) The car now drives along a circular vertical track as shown. The radius of the vertical circle is 9 m. At point A, the car is driving at 30 m/s. Calculate the normal force felt the driver of the car at this point if the driver has a mass of 60 kg.

